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Terminal Offshore Loading Line Intelligent Pigging

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Abstract

This paper undertakes Terminal offshore loading intelligent pigging operation, refers to the use of intelligent (smart) pigs to inspect and maintain subsea pipelines that transport Murban oil between an onshore terminal and SPM (Single Point Mooring). Offshore loading lines are critical assets—any failure can lead to huge environmental, safety, and economic consequences. Intelligent pigging allows non-destructive testing without stopping operations for long. Regulatory bodies (like API, DNV, ISO) often require periodic inspection of subsea pipelines.

In-line-inspection (ILI Pigging) technique Ideal for long subsea export pipelines and as the alternative of Umbilical pigging (tethered tool) technique.

The primary objective of this paper is to propose practical and comprehensive mitigation strategy, which is designed to minimise the incomplete in line inspection due to various reasons. Thus, enabling safety, data accuracy, time efficiency, overall cost effective in terms of production lost time.

Introduction

World-wide oil and gas energy segment is under the transformation of new technology and innovations to unlock and maximizing the value in parallel with long term preservation of facility.

Due to the cost optimisation the overall production lost time / delays are ignored or not given required attention, however, use of ILI technique is preventive, intelligent, and digitally integrated pigging, with an emphasis on subsea-friendly launch/receive systems and combining cleaning with inspection in the same run.

An asset integrity management program aims to determine the condition of a terminal export pipeline and subsea-line till SPM (single point Mooring) and the maintenance required to avoid a critical failure of the asset using non-destructive testing (NDT) methods. In-line inspection (ILI) tools, referred to as intelligent pipeline integrity gauges (PIGs) or smart PIGs, carry a variety of sensors to inspect pipelines for internal and external corrosion, geometric deformation, lamination, cracks, and other defects.

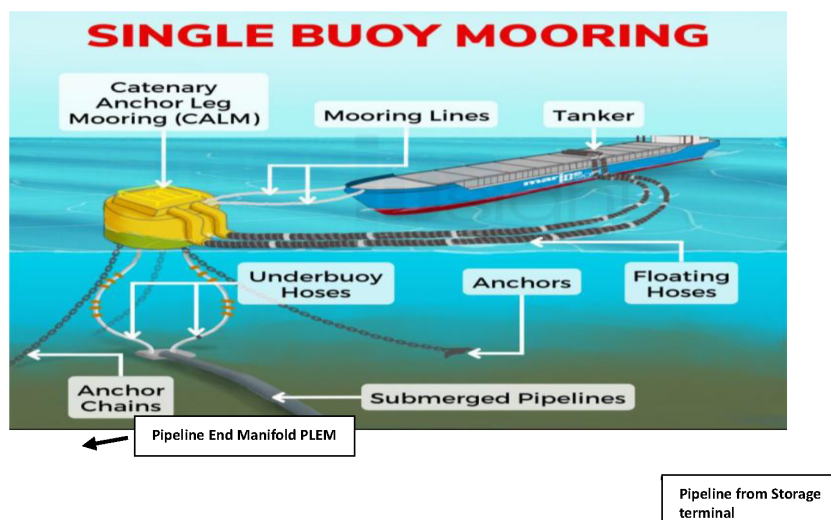


Figure 1—For reference – SPM/CALM facility for oil export by Vessels.

Explanatory note on pigging

Pigs are free flowing objects in pipeline, which is why their progress measurement is depending on combined measurement of pressure, time and volume. Pigs have disc for supporting them in the pipe centre and they move by sealing the discs against the pipe wall at generally flow velocity by an estimated differential pressure (DP). If in any case DP increasing steadily, due to excessive debris, wax contents, operation shall be returned the pig if its alarming.

Another check on the pig performance is to compare the distance travelled against the volume pumped. That will be done by using the fiscal meter and compare the data by pipe volume and the distance to the tracking point. To ensure all facts are correct there will be " pig monitoring and control sheet" for checking exactly all these facts.

Using this system requires good communication, time keeping, record keeping, assessments of the results, this will ensure successful execution of the field activities at the minimum risk level.

Oil tanker requirement

Contractor shall provide an Oil tanker for reverse flow during the pigging operation as per below requirements

1. The tanker shall have standard size of 100,000MT SDWT required for reverse flow.
2. The tanker shall have minimum of three COP's with the capability of pumping Murban crude with discharge rate of 2500-3000 m³/Hr
3. The tanker shall have stripping facility to strip the tanks completely without leaving any cargo after pigging operation.
4. The tanker shall have a fully operational IGS capable of inerting and maintaining the COT's to less than 5% O₂ content, she shall fully comply with her class requirements, international regulations.

Flow control sheet

Tanker In order to establish safe operations and ensure tool position is known at any point of time, flow control sheet is prepared reflecting the required flow at all times during pigging operations. Below tracking points are pre-set and tracking team will be stationed at each point to confirm tool passage time;

1. Point-1 is at flange downstream UT flow meter

2. Point-2 is at approx. 850m downstream the flange
3. Point-3 is at the golden joint
4. Point-4 is 150m upstream the PLEM
5. All pigs will be pushed with an initial flow rate of 20000bbl/hr.
6. Flow rate should be reduced from 20000bb/hr to 10000bbl/hr at 1000m upstream point-4.

Profile Pilot Run

The pigging activities will begin with the pipeline proving run to ensure safe passage of the cleaning and inspection tools. The profile tool run will be done using a Bi-Di profile tool equipped with tracker transmitter, one gauge and one bend plate sized to confirm passage in 85% ID clearance. The profile run will be made up to Point-3 as above (golden joint), where contractors rep will be stationed to record the actual time of tool arrival as well as record any observations on the required time for the tool to stop moving pumping is stopped. Required volume to fill the pipeline up to the golden joint must be pumped exactly. The actual volume pumped shall be checked by terminal fiscal meter as well as contractors UT flow meter. The profile tool will be pumped back to the trap. After trap isolation and draining, the tool conditions as well as the amount of debris "if any" will be checked. The debris would be settled in that low level location and assessing that quantity gives us an impression to that effect.

Profile Run (Gauge run)

Upon successful profile pilot run, the profile tool will be launched and run all the way until 150m before the PLEM. Tracking devices as well as flow calculations shall be used to ensure the tool position is monitored while travelling. As the pipeline is exposed at 150m from the PLEM, company's diver will be required to confirm the tool position. This will establish the confidence in the tracking devices as well as in the calculations of the required volume to be pumped during following runs. The tool will be then pumped back to the traps, where the tool conditions will be checked to confirm the minimum ID of the pipeline. The gauge and bend plates shall be cleaned and measured for any damage. In the event that the gauge and bend plates show any damage, an analysis of the results and conclusions will be made before proceedings with further runs.

Cleaning pigs

The data quality of high resolution in-line inspection tool surveys largely depends on the cleanliness of the pipeline. Accumulation of waxy, ferrous or construction debris may clog the tool, prevent it from rotating properly and damage discs and sensors. A careful progressive cleaning approach shall be adopted to clean the pipeline.

Considerations taken during cleaning runs, (Referring to Fig-2 below)

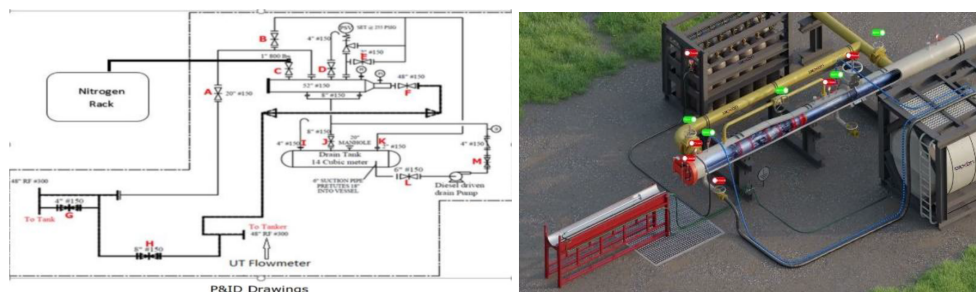


Figure 2—P&ID / arrangement for temporary connections

The drain system in the temporary facility shall be a closed loop to avoid oil spill or blockages due to debris. The cleaning pigs are designed for reverse cleaning action, that is to say, the pig will not remove any debris in the way forward so that no debris ends at the PLEM side. The pig will act as only cleaning tool in its reverse direction. The cleaning tool will be equipped with tracking system. Handling and disposal of all debris from the pipeline will be done by the contractor.

Wax Deposits

Wax Properties	Classifications
HARD = Thick, asphalt properties with 35mm diam hard lumps that require forceful effort to mould or break by hand	5
MEDIUM = Tarry with lumps of a putty consistency 15mm diam mouldable by hand.	4
LIGHT = Pasty consistency some small size lumps 5mm diam that are easily mouldable by fingers.	3
SOFT = Greasy consistency, stick to pig but easily removed by light scrapping.	2
LIQUID = Fluid consistency runs easily from the trap and off pigs.	1

HR-MFL pigging: (Referring to Fig-3 below)



Figure 3—MFL tool

Installation Upon completion of the cleaning tool, Contractor will run its high resolution MFL tool. The flow rate shall be maintained in the range between 0.3 – 1m/s. A trial run will be done where tool will push for 500m from the trap and pumped back. The data collected during trial run will be downloaded and evaluated for the quality; Odometer readings will also be verified for accuracy. Upon acceptance of the trial run results, the tool will be launched for the complete distance of the pipeline. As the MFL tool in its final approach to the PLEM, pumping will be stopped and the tool location is carefully determined by divers. Again the pumped volume during previous runs will be used to monitored and confirm the tool position.

Basic operation procedure: (Referring to Fig-4 below)

The basic operation as below,

1. Installation of pig launcher at terminal side.
2. Final agreement of communication procedure.
3. Test the communication in dry run test.
4. Perform the final leak test on the flange connections of the temporary facility with crude oil.

5. Finalise the berthing of the tanker and connect the tanker.
6. Pump some crude oil to the tanker as a buffer for pumping pigs back to terminal.
7. Perform a test run with the gauge type pig on slightly into offshore (fully monitored and controlled)
8. Run the Gauge pig to 150 before the PLEM and back.
9. Run the cleaning pigs as required.
10. Run the calliper pig if required.
11. Run MFL (magnetic flux leakage) pig to the end (200m before PLEM) and back.
12. Get report of acceptance of data quality and quantity
13. Remove the sludge in holding tank by vacuum truck
14. Complete the final reporting.

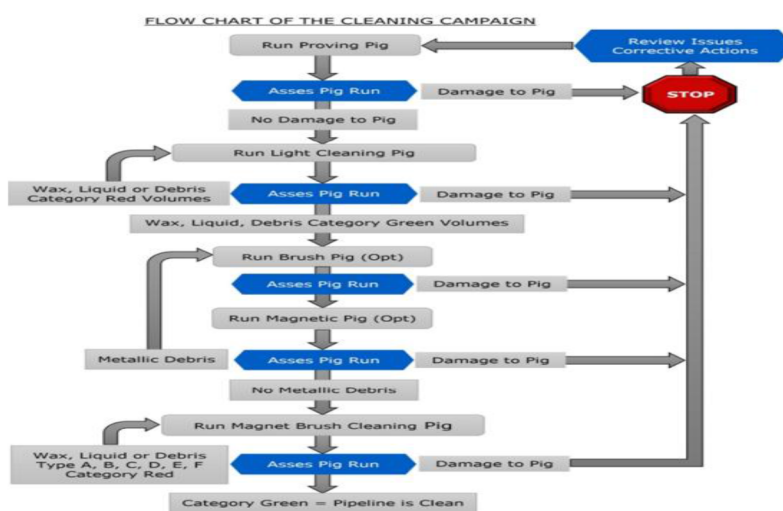


Figure 4—Flow chart for basic operation

XYZ Geo mapping Survey (Referring to Fig-5 below)

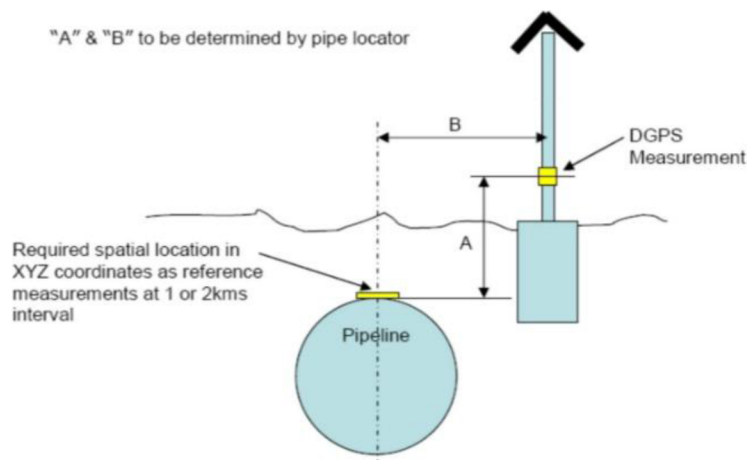


Figure 5

This consist of Geo mapping unit which will provide information regarding 3D route of the pipeline. It has inspection tool and producing digital pipeline routing map information including topographical into GIS. It has DGPS survey of the pipeline route, integrated high DGPS data coordinates on all features along the pipeline including girth wield in the final inspection report. This device will produce accurate routing information in digital data format. The INS XYZ data will be correlated with DGPS reference measurements

at not less than 1km interval (see attached sketch below). The correlated data will finally be integrated digitally in final report giving locations of all reported features with geographical coordinates.

The pipeline routing data will also be presented in topographical format which is very useful for assessing internal anomalies as it shows relationship between low points and possible internal corrosion, if any.

Onshore tracking procedure

Contractor will equip all gauging and cleaning pigs, Bi-Di calliper and Bi-Di MFL tools with low frequency transmitters. The pig locator unit operated by the pig tracking crew will detect them from aboveground. Tool passage time shall be recorded and compared with the calculated time. Contractor will install above ground markers at 0.0km, 0.8km, 1.8km. Upon AGMs placement, contractor will record and submit the details of the marker location with GPS coordinates. This will be required to collect the positioning data.

Offshore tracking procedure (Referring to Fig-6 below)

Below facts are necessary to prepare offshore tracking methods

1. The pumped volume from the terminal is accurately measured using Fiscal meter.
2. The tools cannot travel longer distance in a liquid pipeline than the volume pumped.
3. The tool speed is well controlled by the pumps at the both ends.
4. Except last 178m, all pipelines are buried below sea bed and pipeline route cannot be accurately identified.
5. Accurate information of pipeline length from the "as built" documents.

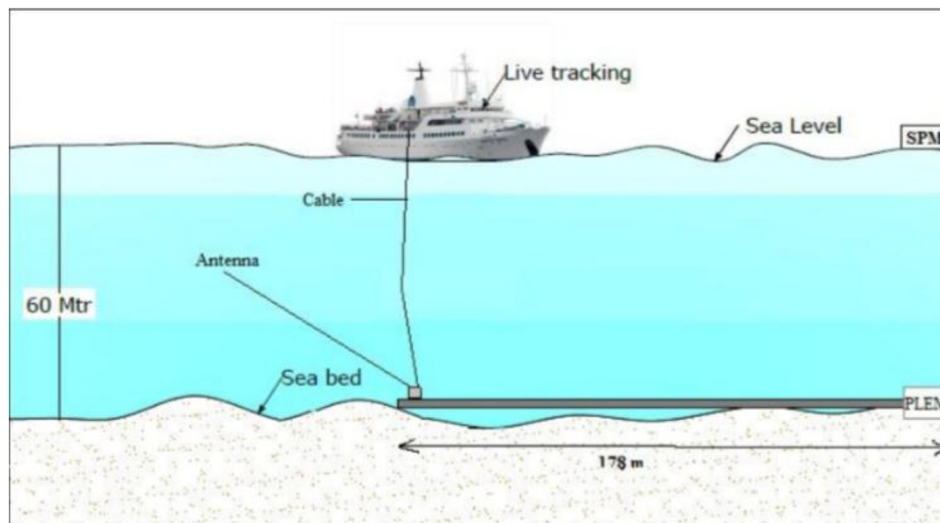


Figure 6—Offshore tracking system sketch

The total length of the longest line is 6851m and bare part at the end is 178m. This means that the last part is about 2.5% of the total length. Monitoring pumped volume will have an error of 0.5%; so when stop pumping at 150m before the end; the pig will have passed the subsea pig tracker at 178m before the end.

When we get the passage signal, appointed diver will be required to dive down with the handheld pig receiver antenna and locate the pig.

1. Shift from Reactive to Preventive Pigging

1. Earlier: Pigging was done mainly when flow performance dropped or a blockage risk appeared.
2. Now: Operators follow scheduled pigging cycles to prevent wax, scale, hydrate, or debris buildup before it affects flow.

3. Reason: Offshore shutdowns are costly, and early intervention is cheaper than repairs or replacement.
2. **Increased Use of Intelligent (ILI) Tools**
 1. Modern preservation programs integrate **In-Line Inspection (ILI) pigs** to combine cleaning with **condition monitoring**:
 - a. Metal loss (corrosion mapping)
 - b. Crack detection
 - c. Geometry anomalies (dents, ovality)
 2. This helps plan repairs **before** failures occur.
3. **Bi-Directional operation & small cargo vessel to Launch/Receive**
 1. In offshore SPM/CBM buoy, normally comes with pipeline end manifold PLEM at the end of the subsea lines, thus based on real-time data calculation and actual length till PLEM, pig travel distance / stoppages decided correctly to avoid overrun and PLEM damage.
4. **Combination Pigging**
 1. Cleaning pigs now often integrate brushes, magnets, and gauging plates in one run.
 2. Multi-pig trains (cleaning pig + batching pig + ILI tool) save vessel time offshore.
5. **Digital Data Integration**
 1. Pigging data is merged into digital twins of the pipeline to predict degradation.
 2. AI-assisted analysis predicts optimal pigging intervals based on flow rate, temperature, wax appearance temperature (WAT), and corrosion monitoring.

Key advantages of ILI over umbilical tool

1. **Full Pipeline Coverage in a Single Run**
 1. ILI tools can traverse the entire pipeline in one continuous run without the length restriction of an umbilical.
 2. Umbilical tools are limited by the physical cable length and deployment setup.
2. **No Need for Heavy Winching or Cable Handling**
 1. ILI is self-propelled by product flow and does not require heavy topside winches or cable reels.
 2. Umbilical pigging needs large winch systems, deck space, and more handling crew.
3. **Ability to Operate in Long, Deepwater Pipelines**
 1. ILI works well for very long offshore export lines (tens to hundreds of kilometres).
 2. Umbilical tools become impractical in deepwater and ultra-long tiebacks due to cable drag and weight.
4. **Reduced Operational Risk**
 1. Fewer moving parts topside and no umbilical tension management → lower risk of tool loss or cable failure.
 2. Umbilical operations carry risk of snagging, twisting, or mechanical failure in the cable.
5. **Better Data Quality & Range of Sensors**
 1. ILI can carry high-resolution MFL, ultrasonic, EMAT, and calliper sensors with precise odometer tracking over the full length.
 2. Umbilical tools are limited by data transmission bandwidth and power supply over long cables.

6. Less Weather Sensitivity Offshore

1. ILI deployment is quicker and requires smaller spreads, so it can be done in shorter weather windows.
2. Umbilical pigging often needs long deck time and stable sea conditions for safe cable handling.

7. Lower Mobilization & Logistics Burden for Remote Locations

1. ILI tools can be launched/received from smaller facilities or FPSOs without massive winching gear.
2. Umbilical systems require more vessel space, heavy equipment, and longer mobilization.

Summary: ILI is the preferred choice for long, deepwater, and continuous inspection with less operational complexity,

1. Operational Efficiency

1. Single-run inspections – ILI tools collect data in one pass, without the need for retrieval cables or multiple trips.
2. No need for vessel station-keeping with a tether – Umbilical pigging limits range due to cable drag and vessel movement; ILI runs freely through long pipelines.
3. Faster deployment & retrieval – Launch and receive from standard pig traps, without long umbilical handling time.

2. Cost Optimization

1. Lower vessel time – Shorter offshore operational windows reduce vessel hire costs.
2. No long umbilical logistics – Eliminates cost and risk of handling heavy cables offshore.
3. Less pipeline downtime – Faster inspection means reduced interruption to export operations.

3. Data Quality & Coverage

1. Full-length inspection – ILI can traverse entire subsea and onshore segments, even beyond umbilical reach limits.
2. High-resolution sensors – Advanced MFL, UT, EMAT, and calliper sensors provide detailed corrosion, dent, and wall loss mapping.
3. Repeatability – Standardized runs allow easier trend analysis over time.

4. Risk Reduction

1. Reduced mechanical stress – No tension from long umbilical pulling on tool.
2. Less entanglement hazard – Eliminates risk of cable snagging in subsea structures.
3. Lower HSE exposure – Less manual handling of heavy cable on deck.

5. Lifecycle Optimization

1. Supports proactive maintenance – Early detection of corrosion or anomalies avoids costly repairs later.
2. Long-term integrity management – Builds a consistent inspection history for regulatory compliance.
3. Future-ready – **ILI platforms are adaptable to new sensor technologies.**

In short – Switching from umbilical pigging to ILI is an optimization because it minimizes time, cost, and operational risks, while maximizing inspection coverage, data accuracy, and long-term asset integrity.

Conclusion

The conclusion of using ILI (In-line Inspection) pigging over the umbilical pigging method in offshore/subsea pipeline integrity work is:

ILI is generally the more efficient, safer, and cost-effective long-term solution because it:

1. Covers the full pipeline in a single run without needing tethering or retrieval via an umbilical, which reduces offshore operational time.
2. Provides higher-quality inspection data (geometry, metal loss, cracks, etc.) in one pass, enabling predictive maintenance and avoiding costly unplanned repairs.
3. Eliminates the need for heavy umbilical and large vessel spreads, lowering logistic complexity, deck footprint, and offshore risk.
4. Reduces downtime since it can be run during normal flow conditions and does not require bidirectional pumping or retrieval setups.
5. Improves safety by avoiding complex subsea interventions and reducing diver/ROV exposure.
6. Supports longer pipeline reach — umbilical tools are limited by cable length and drag, but ILI pigs can inspect pipelines many tens or hundreds of kilometres long.

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Reference

1. Operating and control philosophy document - Main oil terminal
2. Technical standard document.